

Database Management System

RDBMS Programs

1) Consider the following Entities and Relationships Customer (cust_no, cust_name, address, city) Loan (loan_no, loan_amt) Relation between Customer and Loan is Many to Many. Constraint: Primary key, loan_amt should be > 0.

Create a Database in 3NF & write queries for following.

- Find details of all customers whose loan is greater than 10 lakhs.
- List all customers whose name starts with 'ba'.
- List names of all customers in descending order who has taken a loan in Nasik city.
- Display customer details having maximum loan amount.
- Calculate total of all loan amount.

= create table customer

```
(
cust_no number(5) primary key,
cust_name varchar(20),
cust_address varchar(30),
cust_city varchar(30)
);

insert into customer values(101,'balaji','rj road','pune');

insert into customer values(102,'baani','rk road','nashik');

insert into customer values(103,'ram','mg road','nashik');

select * from customer;
```

CUST_NO	CUST_NAME	CUST_ADDRESS	CUST_CITY
101	balaji	rj road	pune
102	baani	rk road	nashik

103

ram

mg road

nashik

create table loan

```
(  
loan_no number(10) primary key,  
loan_amt number(20) constraint chkloan check(loan_amt>0)  
);
```

insert into loan values(1,1200000);

insert into loan values(2,1300000);

insert into loan values(3,110000);

select * from loan;

LOAN_NO	LOAN_AMT
---------	----------

1	1200000
2	1300000
3	110000

create table cust_loan

```
(  
cust_no number(5) references customer(cust_no),  
loan_no number(10) references loan(loan_no)  
);
```

insert into cust_loan values(101,1);

insert into cust_loan values(102,2);

insert into cust_loan values(103,3);

```
select * from cust_loan;
```

CUST_NO LOAN_NO

101	1
102	2
103	3

Q1)find details of all customers whose loan is greater than 10 lakhs

```
--> select cust_name,cust_address,cust_city,loan_amt
from customer,loan,cust_loan
where customer.cust_no=cust_loan.cust_no AND
loan.loan_no=cust_loan.loan_no AND
loan_amt>1000000;
```

CUST_NAME CUST_ADDRESS

CUST_CITY LOAN_AMT

balaji	rj road
pune	1200000
baani	rk road
nashik	1300000

Q2)list all customers whose name start with 'ba'.

```
-->select cust_name from customer where cust_name like 'ba%';
```

CUST_NAME

balaji

baani

Q3)list name of all customers in descending order whos has taken loan in nasik city.

```
--> select cust_name,cust_city
from customer,loan,cust_loan
where customer.cust_no=cust_loan.cust_no AND
loan.loan_no=cust_loan.loan_no AND
cust_city='nashik' order by cust_name desc;
```

CUST_NAME	CUST_CITY
-----	-----
ram	nashik
baani	nashik

Q4)Display customer details having maximum loan amount

```
--> select cust_name,cust_address,cust_city,loan_amt
from customer,loan,cust_loan
where customer.cust_no=cust_loan.cust_no AND
loan.loan_no=cust_loan.loan_no AND
loan_amt=(select MAX(loan_amt)from loan);
```

CUST_NAME	CUST_ADDRESS
-----	-----
CUST_CITY	LOAN_AMT
-----	-----
baani	rk road
nashik	1300000

Q5)Calculate total of all loan amount

```
--> select sum(loan_amt) from loan;
```

```
SUM(LOAN_AMT)
```

```
-----
```

```
2610000
```

2) Consider the following Entities and Relationships Book (Book_no, title, author, price, year_published) Customer (cid, cname, addr) Relation between Book and Customer is Many to Many with quantity as descriptive attribute. Constraint: Primary key, price should be >0. Create a Database in 3NF & write queries for following.

- Display customer details from 'Mumbai'.
- Display author wise details of book.
- Display all customers who have purchased the books published in the year 2013.
- Display customer name that has purchased more than 3 books.
- Display book names having price between 100 and 200 and published in the year 2013.

```
create table book
```

```
(
```

```
Book_no number(5)primary key,
```

```
title varchar(20),
```

```
author varchar(20),
```

```
price number(10) constraint chkbook check(price>0),
```

```
year_published number(5)
```

```
);
```

```
insert into book values(101,'wings of fire','A.P.J.Abdulkalam',400,2014);
```

```
insert into book values(102,'Panipat','Vishwas Patil',150,2013);
```

```
insert into book values(103,'chava','shivaji sawant',200,2013);
```

```
SQL> select * from book;
```

BOOK_NO	TITLE	AUTHOR	PRICE	YEAR_PUBLISHED
101	wings of fire	A.P.J.Abdulkalam	400	2014
102	Panipat	Vishwas Patil	150	2013
103	chava	shivaji sawant	200	2013

create table customer6

(

cust_id number(5) primary key,

cust_name varchar(20),

cust_add varchar(20)

);

insert into customer6 values(201,'Ram','pune');

insert into customer6 values(202,'Sham','Mumbai');

insert into customer6 values(203,'Radha','Mumbai');

SQL> Select * from customer6;

CUST_ID	CUST_NAME	CUST_ADD
201	Ram	pune
202	Sham	Mumbai
203	Radha	Mumbai

create table book_cust

(

book_no number(5) references book(book_no),

cust_id number(5) references customer6(cust_id),

quantity number(5)

);

insert into book_cust values(101,201,2);

insert into book_cust values(102,202,4);

insert into book_cust values(103,203,5);

SQL> select * from book_cust;

BOOK_NO	CUST_ID	QUANTITY
-----	-----	-----
101	201	2
102	202	4
103	203	5

Q1)Display customer details from 'Mumbai'.

SQL> Select * from customer6 where cust_add='Mumbai';

CUST_ID	CUST_NAME	CUST_ADD
-----	-----	-----
202	Sham	Mumbai
203	Radha	Mumbai

Q2)Display author wise details of book.

SQL> Select * from book order by author;

BOOK_NO	TITLE	AUTHOR	PRICE	YEAR_PUBLISHED
-----	-----	-----	-----	-----

101 wings of fire	A.P.J.Abdulkalam	400	2014
102 Panipat	Vishwas Patil	150	2013
103 chava	shivaji sawant	200	2013

Q3)Display all customers who have purchased the books published in the year 2013.

```
--> Select cust_name,year_published

from book,customer6,book_cust

where book.book_no=book_cust.book_no AND

customer6.cust_id=book_cust.cust_id  AND

year_published='2013';
```

CUST_NAME	YEAR_PUBLISHED
-----	-----
Sham	2013
Radha	2013

Q4)Display customer name that has purchased more than 3 books.

```
-->select cust_name,title,quantity

from book,customer6,book_cust

where book.book_no=book_cust.book_no AND

customer6.cust_id=book_cust.cust_id  AND

quantity>3;
```

CUST_NAME	TITLE	QUANTITY
-----	-----	-----
Sham	Panipat	4
Radha	chava	5

Q5)Display book names having price between 100 and 200 and published in the year 2013.


```
-->select title,price from book

where price between 100 AND 200 AND

year_published='2013';
```

TITLE	PRICE

Panipat	150
chava	200

3) Consider the following Entities and Relationships Department (dept_no, dept_name, location) Employee (emp_no, emp_name, address, salary, designation) Relation between Department and Employee is One to Many Constraint: Primary key, salary should be > 0. Create a Database in 3NF & write queries for following.

- Find total salary of all computer department employees.
- Find the name of department whose salary is above 10000.
- Count the number of employees in each department.
- Display the maximum salary of each department.
- Display department wise employee list.

```
create table department

(

dept_no number(5) primary key,

dept_name varchar(20),

dept_location varchar(30)

);

-----

insert values in department table#

insert into department values(101,'computer','pune');

insert into department values(102,'maths','pune');

insert into department values(103,'computer','baramati');
```

```
insert into department values(104,'electronics','baramati');
```

```
select * from department;
```

DEPT_NO	DEPT_NAME	DEPT_LOCATION
101	computer	pune
102	maths	pune
103	computer	baramati
104	electronics	baramati

```
create table employee
(
emp_no number(5) primary key,
emp_name varchar(20),
emp_address varchar(30),
emp_salary number(10) constraint chkemployee check(emp_salary>0),
emp_designation varchar(20),
dept_no number(5) references department(dept_no)
);
```

```
insert into employee values(201,'raj','pune',20000,'professor',101);
insert into employee values(202,'Sham','pune',25000,'professor',102);
insert into employee values(203,'Radha','Baramati',25000,'Assi-professor',103);
insert into employee values(204,'rohan','Baramati',35000,'Assi-professor',104);
```

select *from employee;

EMP_NO	EMP_NAME	EMP_ADDRESS	EMP_SALARY

EMP_DESIGNATION	DEPT_NO		

201 raj	pune	20000	
professor	101		
202 Sham	pune	25000	
professor	102		
203 Radha	Baramati	25000	
Assi-professor	103		
EMP_NO	EMP_NAME	EMP_ADDRESS	EMP_SALARY

EMP_DESIGNATION	DEPT_NO		

204 rohan	Baramati	35000	
Assi-professor	104		

1)Find total salary of all computer department employees.

--> Select sum(emp_salary) as total_salary from department,employee

where department.dept_no=employee.dept_no And

dept_name='computer';

TOTAL_SALARY

45000

2)Find the name of department whose salary is above 10000.

```
-->select dept_name,emp_salary from department,employee
      where department.dept_no=employee.dept_no AND
      emp_salary>10000;
```

DEPT_NAME EMP_SALARY

computer	20000
maths	25000
computer	25000
electronics	35000

3)Count the number of employees in each department.

```
-->select dept_name,count(*) as count from department,employee
      where department.dept_no=employee.dept_no group by dept_name;
```

DEPT_NAME COUNT

maths	1
electronics	1
computer	2

4)Display the maximum salary of each department.

```
-->select max(emp_salary),dept_name from department,employee
```

where department.dept_no=employee.dept_no group by dept_name

MAX(EMP_SALARY) DEPT_NAME

25000	maths
35000	electronics
25000	computer

5)Display department wise employee list.

--> select dept_name,emp_name from department,employee
where department.dept_no=employee.dept_no order by dept_name;

DEPT_NAME	EMP_NAME
computer	raj
computer	Radha
electronics	rohan
maths	Sham

4) Consider the following Entities and Relationships Property (pno, desc, area, rate) Owner (owner_name, addr, phno) Relation between owner and Property is One to Many. Constraint: Primary key, rate should be > 0. Create a Database in 3NF & write queries for following.

- Display area wise property details.
- Display property owned by 'Mr.Patil' having minimum rate.
- Display all properties with owner name that having highest rate of properties located in Chinchwad area.
- Display owner wise property detail.
- Display owner name having maximum no. of properties.

create table property

```
(
    pno number(5) primary key,
    pdesc varchar(20),
    parea varchar(30),
    prate number(10)
);

insert into property values(101,'farm','pune',10000);

insert into property values(102,'house','Chinchwad',10000);

insert into property values(103,'land','Chinchwad',15000);

select * from property;
```

PNO	PDESC	PAREA
-----	-----	-----
PRATE		
101	farm	pune
10000		
102	house	Chinchwad
10000		
103	land	Chinchwad
15000		

```
create table owner
(
    owner_name varchar(20),
    owner_phno number(10),
    owner_address varchar(30),
    pno number(5) references property(pno)
);
```



```
insert into owner values('sham',9433638543,'pune',101);

insert into owner values('mr.patil',7089678786,'pune',102);

insert into owner values('mr.patil',7089678786,'pune',103);
```

1)Display area wise property details.

```
--> select parea,owner_name,pdesc,prate from property,owner

where property.pno=owner.pno order by parea;
```

PAREA	OWNER_NAME	PDESC

PRATE		

Chinchwad	mr.patil	house
10000		
Chinchwad	mr.patil	land
15000		
pune	sham	farm
10000		

2) Display property owned by 'Mr.Patil' having minimum rate.

```
-->select pdesc,parea,prate,owner_name  from property,owner

where property.pno=owner.pno AND

prate=(select min(prate) from property where owner_name='mr.patil');
```

PDESC	PAREA

PRATE	OWNER_NAME

house	Chinchwad
10000	mr.patil

3) Display all properties with owner name that having highest rate of properties located in Chinchwad area.

```
-->select pdesc,parea,prate,owner_name  from property,owner

where property.pno=owner.pno AND

prate=(select max(prate) from property where parea='Chinchwad');
```

PDESC	PAREA

PRATE	OWNER_NAME

land	Chinchwad
15000	mr.patil

4)Display owner wise property detail.

```
--> select owner_name,pdesc,parea,prate from property,owner

where property.pno=owner.pno order by owner_name;
```

OWNER_NAME	PDESC	PAREA

PRATE		

mr.patil	house	Chinchwad
10000		
mr.patil	land	Chinchwad
15000		
sham	farm	pune
10000		

5)Display owner name having maximum no. of properties


```
---> select owner_name,count(*)from property,owner
      where property.pno=owner.pno group by owner_name;
```

OWNER_NAME	COUNT(*)

sham	1
mr.patil	2

5) Consider the following Entities and Relationships Bus (bus_no, capacity, depot_no) Route (rout_no, source, destination, no_of_stations) Relation between Bus and Route is Many to One. Constraint: Primary key. Create a Database in 3NF & write queries for following.

- Find out the route details on which buses whose capacity is 20 runs.
- Display number of stations from 'Chinchwad' to 'Katraj'.
- Display the route on which more than 3 buses runs.
- Display number of buses of route ‘Swargate’ to ‘Hadapsar’.
- Find the bust having maximum capacity from ‘Nigadi’ to 'Kothrud'.

```
create table route
(
rout_no number(5) primary key,
source varchar(20),
destination varchar(30),
no_of_station number(10)
);

insert into route values(3,'swargate','hadpsar',3);

insert into route values(2,'chinchwad','katraj',6);

insert into route values(4,'nigadi','kothrud',7);

select * from route;
```

ROUT_NO	SOURCE	DESTINATION	NO_OF_STATION
3	swargate	hadpsar	3
2	chinchwad	katraj	6
4	nigadi	kothrud	7

create table bus

(
bus_no number primary key,
capacity number(20),
depot_no number(10),
rout_no number(5) references route(rout_no)
);

insert into bus values(201,20,101,3);

insert into bus values(202,21,102,2);

insert into bus values(203,25,103,4);

insert into bus values(204,25,103,4);

insert into bus values(205,27,103,4);

insert into bus values(206,26,103,4);

select * from bus;

BUS_NO	CAPACITY	DEPOT_NO	ROUT_NO
201	20	101	3
202	21	102	2
203	25	103	4
204	25	103	4
205	27	103	4

1)Find out the route details on which buses whose capacity is 20 runs.

```
-->select * from bus,route where route.rout_no=bus.rout_no and capacity=20;

BUS_NO  CAPACITY  DEPOT_NO  ROUT_NO  ROUT_NO SOURCE
```

DESTINATION NO_OF_STATION

201	20	101	3	3	swargate
hadpsar			3		

2)Display number of stations from 'Chinchwad' to ' Katraj'.

```
-->select no_of_station from route where source='chinchwad' and destination='katraj';
```

NO_OF_STATION

6

3)Display the route on which more than 3 buses runs.

```
--> select route.rout_no,source,destination from bus,route where route.rout_no=bus.rout_no group by
route.rout_no,source,destination
having count(bus.rout_no)>3;
```

ROUT_NO SOURCE DESTINATION

4	nigadi	kothrud
---	--------	---------

4)Display number of buses of route ‘Swargate’ to ‘Hadapsar’.

```
--> select count(bus_no) from bus,route where route.rout_no=bus.rout_no and source='swargate' and
destination='hadapsar';
```

COUNT(BUS_NO)

1

5) Find the bus having maximum capacity from ‘Nigadi’ to 'Kothrud'..

--> select bus_no,capacity from bus,route where route.rout_no=bus.rout_no and source='nigadi' and destination='kothrud' group by bus_no,capacity

having capacity=(select max(capacity) from bus,route);

BUS_NO CAPACITY

203 25

6) Consider the following Entities and Relationships Person (pno, person_name, birthdate, income) Area (area_name, area_type) Relation between Person and area is Many to One. Constraint: Primary key, income should be > 0, area_type should be rural or urban. Create a Database in 3NF & write queries for following.

- Display persons having income less than 1 lakhs in PCMC Area.
- Display population of each area.
- Display person’s details from 'Urban' area.
- Display the details of area having population greater than that of in Pune.
- Display details of person from each area having minimum income.

create table area1

(

a_no number(5) primary key,

area_name varchar(20),

area_type varchar(20),

population number(10)

);

```
insert into area1 values(101,'pune','urban',4000);

insert into area1 values(102,'baramati','rural',5000);

insert into area1 values(103,'mumbai','urban',6000);

insert into area1 values(104,'pcmc','urban',10000);

select * from area1;
```

A_NO	AREA_NAME	AREA_TYPE	POPULATION
101	pune	urban	4000
102	baramati	rual	5000
103	mumbai	urban	6000
104	pcmc	urban	10000

```
create table person1

(

p_no number(5) primary key,

p_name varchar(20),

p_birthdate date,

p_income number(10) constraint chkperson1 check(p_income>0),

a_no number(5) references area1(a_no)

);
```

```
insert into person1 values(201,'ram','1-jan-2001',10000,101);

insert into person1 values(202,'sham','14-july-1998',15000,102);

insert into person1 values(203,'radha','24-nov-1997',20000,103);

insert into person1 values(204,'dev','27-dec-1996',25000,104);

select * from person1;
```

P_NO	P_NAME	P_BIRTHDA	P_INCOME	A_NO
201	ram	01-JAN-01	10000	101
202	sham	14-JUL-98	15000	102
203	radha	24-NOV-97	20000	103
204	dev	27-DEC-96	25000	104

1)Display persons having income less than 1 lakhs in PCMC Area.

```
-->select p_name from area1,person1 where area1.a_no=person1.a_no AND  
area_name='pcmc' AND p_income<100000;
```

P_NAME

dev

2)Display population of each area.

```
-->select population,area_name from area1,person1 where area1.a_no=person1.a_no group by  
population,area_name;
```

POPULATION AREA_NAME

6000 mumbai
4000 pune
5000 baramati
10000 pcmc

3)Display person’s details from 'Urban' area.

```
-->select p_name from area1,person1 where area1.a_no=person1.a_no AND area_type='urban';
```

P_NAME

ram

radha

dev

4)Display the details of area having population greater than that of in Pune.

-->select max(population),area_name from area1 where area_name='pune' group by area_name;

MAX(POPULATION) AREA_NAME

4000 pune

5)Display details of person from each area having minimum income

-->select p_name,p_birthdate,p_income,area_name from area1,person1

where area1.a_no=person1.a_no And p_income=(select min(p_income) from person1);

P_NAME P_BIRTHDA P_INCOME AREA_NAME

ram 01-JAN-01 10000 pune

7) Consider the following Entities and Relationships. Room (roomno, desc, rate) Guest (gno, gname, no_of_days) Relation between Room and Guest is One to One. Constraint: Primary key, no of days should be > 0. Create a Database in 3NF & write queries for following.

- Display room details according to its rates in ascending order.
- Find the names of guest who has allocated room for more than 3 days.
- Find no. of AC rooms.
- Display total amount for NON-AC rooms.
- Find names of guest with maximum room charges.

create table room1

(

```
r_no number(5)primary key,  
  
r_desc varchar(20),  
  
r_rate number(10)  
  
);  
  
insert into room1 values(101,'AC',5000);  
  
insert into room1 values(102,'NON AC',3000);  
  
insert into room1 values(103,'AC',8000);  
  
insert into room1 values(104,'NON AC',4000);  
  
select * from room1;
```

R_NO	R_DESC	R_RATE
-----	-----	-----
101	AC	5000
102	NON AC	3000
103	AC	8000
104	NON AC	4000

```
create table guest1  
  
(  
  
g_no number(5) primary key,  
  
g_name varchar(20),  
  
no_of_days number(5) constraint chkguest1 check(no_of_days>0),  
  
r_no number(5)unique references room1(r_no)  
  
);  
  
  
insert into guest1 values(201,'ram',2,101);  
  
insert into guest1 values(202,'sham',3,102);  
  
insert into guest1 values(203,'radha',4,103);  
  
insert into guest1 values(204,'dev',6,104);
```



```
select * from guest1;
```

G_NO	G_NAME	NO_OF_DAYS	R_NO
201	ram	2	101
202	sham	3	102
203	radha	4	103
204	dev	6	104

1)Display room details according to its rates in ascending order.

```
select * from room1 order by r_rate asc;
```

R_NO	R_DESC	R_RATE
102	NON AC	3000
104	NON AC	4000
101	AC	5000
103	AC	8000

2)Find the names of guest who has allocated room for more than 3 days.

```
select g_name from guest1 where no_of_days>3;
```

G_NAME
radha
dev

3) Find no. of AC rooms.

```
select count(r_no) as count from room1 where r_desc='AC';
```

COUNT

2

4)Display total amount for NON-AC rooms.

select count(r_no) as count from room1 where r_desc='NON AC';

COUNT

2

5)Find names of guest with maximum room charges

select g_name from guest1,room1 where guest1.r_no=room1.r_no and

r_rate=(select max(r_rate) from room1);

G_NAME

Radha

8) Consider the following Entities and Relationships Branch (bname , bcity ,assets) Loan (loan_no, amount)
Relation between Branch and Loan is one to many. Constraint: Primary key, amount and assets should be > 0.

Create a Database in 3NF & write queries for following.

- Display total loan amount given by DYP branch.
- Find total number of loans given by each branch.
- Find the name of branch that have maximum assets located in Mumbai.
- Display loan details in descending order of their amount.
- Display all branches located in Mumbai, Pune and Nasik.

create table branch1

```
(
b_no number(5)primary key,
b_name varchar(20),
b_city varchar(20),
assets number(5) constraint chkbranch1 check(assets>0)
);
```

```
insert into branch1 values(101,'BOI','pune',38000);
insert into branch1 values(102,'DYP','mumbai',34090);
insert into branch1 values(103,'SBI','pune',38180);
insert into branch1 values(104,'UNION','mumbai',40000);
select * from branch1;
```

B_NO	B_NAME	B_CITY	ASSETS
101	BOI	pune	38000
102	DYP	mumbai	34090
103	SBI	pune	38180
104	UNION	mumbai	40000

```
create table loan1
(
loan_no number(10) primary key,
loan_amount number(10),
b_no number(5) references branch1(b_no)
);
```

```
insert into loan1 values(201,40000,101);
insert into loan1 values(202,27000,102);
```

```
insert into loan1 values(203,38000,103);
```

```
insert into loan1 values(204,44000,104);
```

```
select * from loan1;
```

LOAN_NO	LOAN_AMOUNT	B_NO
-----	-----	-----
201	40000	101
202	27000	102
203	38000	103
204	44000	104

1)Display total loan amount given by DYP branch.

```
-->select loan_amount,b_name from branch1,loan1 where branch1.b_no=loan1.b_no AND  
b_name='DYP';
```

LOAN_AMOUNT	B_NAME
-----	-----
27000	DYP

2)Find total number of loans given by each branch.

```
-->select b_name,count(*) from branch1,loan1 where branch1.b_no=loan1.b_no group by b_name;
```

B_NAME	COUNT(*)
-----	-----
SBI	1
BOI	1
DYP	1
UNION	1

3)Find the name of branch that have maximum assets located in Mumbai.

-->Select b_name,b_city from branch1, loan1

Where branch1.b_no=loan1.b_no and

assets=(select max(assets) from branch1 where

b_city='mumbai');

B_NAME B_CITY

UNION mumbai

4)Display loan details in descending order of their amount.

-->select loan1.b_no,b_name,b_city,assets,loan_amount from branch1,loan1

where branch1.b_no=loan1.b_no order by loan_amount desc;

B_NO B_NAME B_CITY ASSETS LOAN_AMOUNT

104 UNION mumbai 40000 44000

101 BOI pune 38000 40000

103 SBI pune 38180 38000

102 DYP mumbai 34090 27000

5)Display all branches located in Mumbai, Pune and Nasik.

-->select b_name,b_city from branch1,loan1

where branch1.b_no=loan1.b_no and

b_city in ('mumbai','pune','nashik');

B_NAME B_CITY

BOI pune

DYP mumbai

SBI pune

UNION Mumbai

PL/SQL Programs

1) Consider the following entities and their relationship. Customer (c_no, c_name, c_city, c_ph_no)

Ticket (t_no, booking_date, fare, traveling_date) Relationship between Customer and Ticket is one-to-many. Constraints: primary key, foreign key c_name should not be null, fare should be greater than zero.

Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1) Write a procedure to display names of customer who have booked bus on given date.

2) Write a trigger that restricts insertion of ticket having traveling date smaller than booking date. (Raise user defined exception and give appropriate message)

create table Customer

(

c_no number(5) primary key,

c_name varchar2(20),

c_city varchar2(20),

phno number(10)

);

insert into Customer values(1,'yogesh','pune',9096188754);

insert into Customer values(2,'Dipali','pune',9096188756);

insert into Customer values(3,'Amol','mumbai',9096188554);

create table Ticket

(

t_no number(5) primary key,

b_date date,


```
fare number(5),  
  
t_date date,  
  
c_no number(5)references Customer(c_no)  
  
);
```

```
-----  
  
insert into Ticket values(1,'02-july-2016',600,'06-july-2016',1);  
  
insert into Ticket values(2,'02-july-2016',600,'09-july-2016',2);  
  
insert into Ticket values(3,'02-july-2016',600,'05-july-2016',3);  
  
insert into Ticket values(4,'06-aug-2016',600,'06-sep-2016',1);
```

```
-----create procedure-----
```

```
set serveroutput on;  
  
create or replace procedure p1(dt in date)  
  
as  
  
cursor c1 is select c.c_name from Customer c, Ticket t  
  
where c.c_no=t.c_no and b_date=dt;  
  
begin  
  
for x in c1 loop  
  
dbms_output.put_line(x.c_name);  
  
end loop;  
  
end;  
  
/
```

Procedure created.

```
-----Call To Procedure-----
```

```
declare  
  
begin  
  
p1('02-july-2016');  
  
end;  
  
/
```

-----Output for procedure-----

yogesh

Dipali

Amol

PL/SQL procedure successfully completed.

-----create trigger-----

set serveroutput on;

create or replace trigger t1

before insert or update on Ticket

for each row

begin

if(:new.b_date>:new.t_date) then

raise_application_error(-20343,'the booking date should be less then traveling date');

end if;

end;

/

Trigger created.

-----output of trigger-----

insert into ticket values(5,'06-JUL-2017',600,'06-May-2017',2);

insert into ticket values(5,'06-JUL-2017',600,'06-May-2017',2)

*

ERROR at line 1:

ORA-20343: the booking date should be less then traveling date

Slip NO-2) Consider the following entities and their relationship.

Student (s_reg_no, s_name, s_class) Competition (comp_no, comp_name, comp_type)

Relationship between Student and Competition is many-to-many with descriptive attribute rank and year.
Constraints: primary key, foreign key, primary key for third table(s_reg_no, comp_no, year), s_name and comp_name should not be null, comp_type can be sports or academic. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1)Write a function which will accept s_reg_no of student and returns total number of competition in which student has participated in a given year.

2)Write a cursor which will display year wise details of competitions held. (Use parameterized cursor

create table Student

(

s_reg_no number(5) primary key,

s_name varchar(20) not null,

s_class varchar(20)

);

-----Value Insertion-----

insert into Student values(1,'raj','f.y.bca');

insert into Student values(2,'rajesh','s.y.bca');

insert into Student values(3,'amol','t.y.bca');

create table Competition

(

comp_no number(5) primary key,

comp_name varchar(20),

comp_type varchar(20)

);

insert into Competition values(101,'cricket','sports');

insert into Competition values(102,'tennis','sports');

insert into Competition values(103,'IPL','academic');

insert into Competition values(104,'football','academic');

```
create table stud_comp
(
s_reg_no number(5) references Student(s_reg_no),
comp_no number(5) references Competition(comp_no),
rank number(5),
year number(5)
);
```

```
insert into stud_comp values(1,101,1,2016);
```

```
insert into stud_comp values(1,102,1,2016);
```

```
insert into stud_comp values(1,103,1,2016);
```

```
insert into stud_comp values(1,104,1,2016);
```

```
insert into stud_comp values(2,101,2,2016);
```

```
insert into stud_comp values(3,101,2,2016);
```

-----Function Defination-----

```
set serveroutput on;
```

```
create or replace function fun1(sregno in number, yr in number)
```

```
    return number
```

```
    is
```

```
    total number;
```

```
begin
```

```
select count(c.comp_no) into total from Student s, Competition c, stud_comp sc
```

```
where s.s_reg_no=sc.s_reg_no and c.comp_no=sc.comp_no and s.s_reg_no=sregno
```

```
and year=yr;
```

```
return total;
```

```
end;
```

```
/
```

Function created.

-----Output For Function-----

```
SELECT fun1(1,2016) from dual;
```

```
FUN1(1,2016)
```

```
-----
```

```
4
```

```
-----Curser-Defination-----
```

```
set serveroutput on;
```

```
declare
```

```
cursor c1(name Competition.comp_type%type)
```

```
is select comp_name,comp_type,year
```

```
from
```

```
Competition c,stud_comp sc
```

```
where c.comp_no=sc.comp_no
```

```
order by year;
```

```
res c1%rowtype;
```

```
v_name Competition.comp_type%type;
```

```
begin
```

```
for res in c1(&v_name)
```

```
loop
```

```
dbms_output.put_line(res.comp_name||' '||res.comp_type||' '||res.year);
```

```
end loop;
```

```
end;
```

```
/-----Output-----
```

```
Enter value for v_name: 'IPL'
```

```
old 11: for res in c1(&v_name)
```

```
new 11: for res in c1('IPL')
```

```
PL/SQL procedure successfully completed.
```

Slip NO-3)Consider the following entities and their relationship. Owner (o_no, o_name, o_city, o_ph_no)

Estate (e_no, e_type, e_city, e_price) Relationship between Owner and Estate is one-to-many. Constraints: primary key, foreign key, o_name should not be null, e_type can be flat, bungalow or land Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1) Write a procedure which will accept owner number and display details of all estates of given owner which belongs to pune city.

2) Write a cursor which will display type wise estate details. (Use parameterized cursor

```
create table Owner
```

```
(
```

```
o_no number(5) primary key,
```

```
o_name varchar(20),
```

```
o_city varchar(20),
```

```
phno number(10)
```

```
);
```

```
insert into Owner values(1,'yogesh','pune',9096188754);
```

```
insert into Owner values(2,'dipali','pune',9096188755);
```

```
insert into Owner values(3,'raj','pune',9096188756);
```

```
create table Estate
```

```
(
```

```
e_no number(5) primary key,
```

```
e_type varchar(20),
```

```
e_city varchar(20),
```

```
e_price number(10),
```

```
o_no number(5) references Owner(o_no)
```

```
);
```

```
insert into Estate values(101,'flat','pune',4000000,1);
```

```
insert into Estate values(102,'land','pune',4500000,1);
```

```
insert into Estate values(103,'bungalow','mumbai',4100000,1);
```

```
insert into Estate values(104,'land','nashik',4900000,2);
```

```
-----Creating Procedure-----
```

```
set serveroutput on;
```

```
create or replace procedure p7(wn in number)
```

```
as
```

```
cursor c1 is select e.e_no,e.e_type,e.e_city,e.e_price from Owner o ,Estate e
```

```
where o.o_no=e.o_no and o.o_city='pune' and o.o_no=wn;
```

```
begin
```

```
for x in c1 loop
```

```
dbms_output.put_line(x.e_no||' '||x.e_type||' '||x.e_city||' '||x.e_price);
```

```
end loop;
```

```
end;
```

```
/
```

```
procedure created
```

```
-----
```

```
declare
```

```
begin
```

```
p7(1);
```

```
end;
```

```
/
```

```
-----Output for procedure-----
```

```
101 flat pune 4000000
```

```
102 land pune 4500000
```

```
103 bungalow mumbai 4100000
```

```
PL/SQL procedure successfully completed.
```

-----Creating Curser-----

```
set serveroutput on;

declare

cursor c1(name Estate.e_type%type)

is select e_no,e_type,e_city,e_price

from Estate

where e_type=name;

res c1%rowtype;

v_name Estate.e_type%type;

begin

for res in c1(&v_name)

loop

dbms_output.put_line(res.e_no||' '||res.e_type||' '||res.e_city||' '||res.e_price);

end loop;

end;

/
```

-----Output For Curser-----

Enter value for v_name: 'land'

old 9: for res in c1(&v_name)

new 9: for res in c1('land')

102 land pune 4500000

104 land nashik 4900000

PL/SQL procedure successfully completed.

slip no-4) Consider the following entities and their relationship. Route(route_no, source, destination, no_of_station) Bus (bus_no, capacity, depot_name) Relationship between Route and Bus is one-to-many Constraints: primary key, foreign key, depot_name should not be null, bus capacity should be greater than 40. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1) Write a procedure which will display all bus details for a given route.

2) Write a trigger that restricts insertion of route having number of station less than zero.

(Raise user defined exception and give appropriate message)

create table Route

(

route_no number(5) primary key,

source varchar(20),

destination varchar(20),

no_of_station number(5)

);

insert into Route values(1,'chikhali','akurdi',3);

insert into Route values(2,'chikhali','nigdi',4);

insert into Route values(3,'akurdi','pune',10);

create table Bus

(

bus_no number(5) primary key,

capacity number(5) check(capacity>40),

depot_name varchar(20),

route_no number(5) references Route(route_no)

);

insert into Bus values(101,43,'nigdi',1);

insert into Bus values(102,43,'chikhali',1);

insert into Bus values(103,43,'nigdi',2);

insert into Bus values(104,43,'pimpri',1);

-----creating procedure-----

set serveroutput on;

create or replace procedure p7(rt in number)

as

cursor c1 is select bus_no, capacity, depot_name from Route r ,Bus b

where r.route_no=b.route_no and r.route_no=rt;

begin

for x in c1 loop

dbms_output.put_line(x.bus_no||' '||x.capacity||' '||x.depot_name);

end loop;

end;

/

Procedure created.

-----call for procedure-----

declare

begin

p7(1);

end;

/

-----output for procedure-----

101 43 nigdi

102 43 chikhali

104 43 pimpri

PL/SQL procedure successfully completed.

-----createing trigger-----

set serveroutput on;

create or replace trigger t11

before insert or update on Route

for each row

begin

if(:new.no_of_station<0) then

raise_application_error(-20343,'the no of station should be greater than zero');

end if;

end;

/

Trigger created.

-----output for trigger-----

insert into route values(4,'Pimpri','swarget',-9);

insert into route values(4,'Pimpri','swarget',-9)

*

ERROR at line 1:

ORA-20343: the no of station should be greater than zero

slip no-5) Consider the following entities and their relationship. Patient (p_no, p_name, p_addr) Doctor (d_no, d_name, d_addr, city) Relationship between Patient and Doctor is many-to-many with descriptive attribute disease and no_of_visits. Constraints: primary key, foreign key, Primary key for third table(p_no, d_no, disease), p_name and d_name should not be null. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1)Write a procedure which will display patient detail who has visited more than 3 times to the given doctor for ‘Diabetes’.

2) Write a trigger which will restrict insertion or updation of doctor_patient details having no_of_visits less than zero. (Raise user defined exception and give appropriate message

create table Patient

(

```
p_no number(5) primary key,
```

```
p_name varchar(20),
```

```
p_addr varchar(20)
```

```
);
```

```
insert into Patient values(1,'raj','pune');
```

```
insert into Patient values(2,'rajesh','pune');
```

```
insert into Patient values(3,'mukesh','mumbai');
```

```
create table Doctor
```

```
(
```

```
d_no number(5) primary key,
```

```
d_name varchar(20),
```

```
d_addr varchar(20),
```

```
city varchar2(20)
```

```
);
```

```
insert into Doctor values(101,'yogesh','pune','pune');
```

```
insert into Doctor values(102,'dipali','pune','pune');
```

```
insert into Doctor values(103,'amol','varsova','mumbai');
```

```
create table P_D
```

```
(
```

```
p_no number(5) references Patient(p_no),
```

```
d_no number(5) references Doctor(d_no),
```

```
disease varchar(20),
```

```
no_of_visits number(5)
```

```
);
```

```
insert into P_D values(1,101,'swineflue',7);
```

```
insert into P_D values(1,101,'dehydration ',7);
```

```
insert into P_D values(1,102,'dehydration ',3);
```

```
insert into P_D values(2,102,'diabetes ',7);
```

```
insert into P_D values(3,103,'diabetes ',2);
```

```
insert into P_D values(1,101,'diabetes',4);
```

```
insert into P_D values(1,101,'asthma',4);
```

```
insert into P_D values(2,101,'asthma',4);
```

```
insert into P_D values(2,101,'diabetes',5);
```

```
insert into P_D values(3,101,'diabetes',6);
```

-----creating procedure-----

```
set serveroutput on;
```

```
create or replace procedure p1(s_name in varchar2)
```

```
as
```

```
cursor c1 is select p.* from Patient p,Doctor d,P_D pd
```

```
where p.p_no=pd.p_no and d.d_no=pd.d_no
```

```
and pd.disease='diabetes' and d.d_name=s_name and pd.no_of_visits>3;
```

```
begin
```

```
for abc in c1
```

```
loop
```

```
dbms_output.put_line('patient no:='||abc.p_no||'patient name:=' ||abc.p_name|| 'patient  
address:='||abc.p_addr);
```

```
end loop;
```

```
end;
```

```
/
```

```
Procedure created.
```

-----call for procedure-----

```
declare
```

```
begin
```

```
p1('yogesh');
```

```
end;
```

```
/
```

```
-----output for procedure-----
```

```
patient no:=1patient name:=rajpatient address:=pune
```

```
patient no:=2patient name:=rajeshpatient address:=pune
```

```
patient no:=3patient name:=mukeshpatient address:=mumbai
```

```
PL/SQL procedure successfully completed.
```

```
-----creating trigger-----
```

```
set serveroutput on;
```

```
create or replace trigger trg
```

```
before insert or update on P_D
```

```
for each row
```

```
begin
```

```
if:new.no_of_visits<0 then
```

```
raise_application_error(-2010,'no_of_visits less than zero');
```

```
end if;
```

```
end;
```

```
/
```

```
Trigger created.
```

```
-----output for trigger-----
```

```
insert into p_d values(4,101,'asthma',-2);
```

```
insert into p_d values(4,101,'asthma',-2)
```

```
      *
```

```
ERROR at line 1:
```

```
ORA-20343: no_of_visits less than zero
```

Slip N0-6) Consider the following entities and their relationship.Crop (c_no, c_name, c_season, pesticides)
Farmer (f_no, f_name, f_location) Relationship between Crop and Farmer is many-to-many with descriptive attribute year. Constraints: primary key, foreign key, primary key for third table(c_no, f_no, year), c_name and f_name should not be null, c_season can be rabi or kharif. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

- 1)Write a function which will return total number of farmers harvesting given crop in a given year.
- 2)write a cursor which will display season wise information of crops harvested by the farmers. (Use parameterized cursor)

```
-----  
  
create table Crop  
  
(  
  
c_no number(5) primary key,  
  
c_name varchar(20)not null,  
  
c_season varchar(20),  
  
pesticides varchar(20)  
  
);
```

```
-----  
  
create table Farmer  
  
(  
  
f_no number(5) primary key,  
  
f_name varchar(20)not null,  
  
f_location varchar(20)  
  
);
```

```
-----  
  
create table C_F  
  
(  
  
c_no number(5) references Crop(c_no),  
  
f_no number(5) references Farmer(f_no),  
  
year number(5)
```

);

insert into Crop values(1,'wheat','rabbi','selfer');

insert into Crop values(2,'rise','rabbi','co2');

insert into Crop values(3,'mung','kharip','calcium');

insert into Crop values(4,'jowar','kharip','phospet');

insert into Farmer values(101,'ram','pune');

insert into Farmer values(102,'ganesh','pune');

insert into Farmer values(103,'rajesh','nashik');

insert into C_F values(1,101,2014);

insert into C_F values(1,102,2014);

insert into C_F values(1,103,2014);

insert into C_F values(2,101,2015);

insert into C_F values(3,101,2015);

insert into C_F values(4,103,2016);

-----function creation-----

set serveroutput on;

create or replace function fun1(yr in number)

return number

is

total number;

select count(f.f_no) into total from Crop c ,Farmer f ,C_F cf

where c.c_no=cf.c_no and f.f_no=cf.f_no and cf.year=yr;

return total;

end;

/

Function created.

-----Output for function-----

```
select fun1(2014) from dual;
```

FUN1(2014)

3

-----cursor creation-----

```
set serveroutput on;
```

```
declare
```

```
cursor c1 is select distinct c_season from Crop;
```

```
cursor c2 (se Crop.c_season%type) is select distinct c.c_no , c.c_name ,c.pesticides,f.f_name
```

```
from Crop c ,Farmer f ,C_F cf
```

```
where c.c_no=cf.c_no and f.f_no=cf.f_no and c.c_season=se;
```

```
begin
```

```
for a in c1 loop
```

```
dbms_output.put_line(a.c_season);
```

```
for c in c2 (a.c_season) loop
```

```
dbms_output.put_line('crop no:='||c.c_no||'crop name:=' ||c.c_name|| 'pesticides:='||c.pesticides||'farmer  
name:='||c.f_name);
```

```
end loop;
```

```
end loop;
```

```
end;
```

/

-----output for cursor-----

kharip

crop no:=4crop name:=jowarpesticides:=phospetfarmer name:=rajesh

```
crop no:=3crop name:=mungpesticides:=calciumfarmer name:=ram
rabbi

crop no:=1crop name:=wheatpesticides:=selferfarmer name:=ram

crop no:=1crop name:=wheatpesticides:=selferfarmer name:=ganesh

crop no:=1crop name:=wheatpesticides:=selferfarmer name:=rajesh

crop no:=2crop name:=risepesticides:=co2farmer name:=ram

PL/SQL procedure successfully completed.
```

Slip No-7) Consider the following entities and their relationship.Customer (c_no, c_name, c_city, c_ph_no)
Ticket (t_no, booking_date, fare, traveling_date) Relationship between Customer and Ticket is one-to-many. Constraints: primary key, foreign key, c_name should not be null, fare should be greater than zero.
Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

- 1) Write a function which will calculate and return total fare collected from customers on given date.
- 2)Write a cursor which will display date wise ticket booked by customer. (Use parameterized cursor).

```
-----

create table Customer11

(

c_no number(5) primary key,

c_name varchar(20),

c_city varchar(20),

phno number(10)

);

-----
```

```
create table Ticket11

(

t_no number(5) primary key,

b_date date,

fare number(5),
```



```
t_date date,
```

```
c_no number(5) references Customer11(c_no)
```

```
);
```

```
-----  
insert into Customer11 values(1,'yogesh','pune',9096188754);
```

```
insert into Customer11 values(2,'Dipali','pune',9096188756);
```

```
insert into Customer11 values(3,'amol','mumbai',9096188554);  
-----
```

```
insert into Ticket11 values(1,'02-july-2016',600,'06-july-2016',1);
```

```
insert into Ticket11 values(2,'02-july-2016',600,'09-july-2016',2);
```

```
insert into Ticket11 values(3,'02-july-2016',600,'05-july-2016',3);
```

```
insert into Ticket11 values(4,'06-aug-2016',600,'06-sep-2016',1);
```

```
-----function creation-----
```

```
set serveroutput on;
```

```
create or replace function f1(p_date in date)
```

```
return number as
```

```
p_amt number(10);
```

```
begin
```

```
select sum(fare) into p_amt from Ticket where b_date=p_date;
```

```
return p_amt;
```

```
end;
```

```
/
```

```
Function created.
```

```
-----function call-----
```

```
declare
```

```
p_date date;
```

```
amt number(10);
```

```
begin
```

```
dbms_output.put_line('enter the date');

p_date:='&p_date';

amt:=fl(p_date);

dbms_output.put_line('total collection is:'||amt);

end;

/
```

-----output for function-----

Enter value for p_date: 02-JUL-16

old 6: p_date:='&p_date';

new 6: p_date:='02-JUL-16';

enter the date

total collection is:=1200

PL/SQL procedure successfully completed.

-----creation of cursor-----

set serveroutput on;

DECLARE

cursor c1 is select DISTINCT b_date from Ticket;

cursor c2(p_date Ticket.b_date%type) is

select c.* from Customer c,Ticket t

where c.c_no=t.c_no

and t.b_date=p_date;

begin

for x in c1

loop

dbms_output.put_line(x.b_date);

for x1 in c2(x.b_date)

loop

dbms_output.put_line(x1.c_no||x1.c_name);

```
end loop;

end loop;

end;

/
```

-----output for cursor-----

02-JUL-16

2Dipali

3amol

06-AUG-16

1yogesh

2Dipali

PL/SQL procedure successfully completed.

Slip No- 8)Consider the following entities and their relationship.

Train (t_no, t_name) Passenger (p_no, p_name, addr, age) Relationship between Train and Passenger is many-to-many with descriptive attribute date, seat_no and amt. Constraints: primary key, foreign key, primary key for third table (t_no, p_no, date), t_name and p_name should not be null, amt should be greater than zero.

Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

- 1) Write a function which will display train details having maximum passenger for a given date.
- 2)write a cursor which will display date wise train and their passenger details

```
create table Train

(

t_no number(5) primary key,

t_name varchar(20) not null

);
```

```
create table Passenger
```

(

p_no number(5) primary key,

p_name varchar(20) not null,

addr varchar(20),

age number(5)

);

create table T_P

(

t_no number(5) references Train(t_no),

p_no number(5) references Passenger(p_no),

date1 date,

seat_no number(5),

amt number(5)

);

insert into Train values(2,'mumbai Express');

insert into Passenger values(101,'yogesh','pune',25);

insert into Passenger values(102,'raj','pune',15);

insert into Passenger values(103,'amol','mumbai',22);

insert into T_P values(1,101,'01-july-2016',1001,600);

insert into T_P values(1,102,'01-july-2016',1002,600);

insert into T_P values(1,103,'02-july-2016',1003,600);

insert into T_P values(2,101,'07-july-2016',1004,500);

insert into T_P values(1,102,'01-july-2014',1005,790);

-----function creation-----

```
set serveroutput on;
```

```
create or replace function f1
```

```
(dat T_P.date1%type)
```

```
return number is
```

```
num number;
```

```
Begin
```

```
select (tp.t_no) into num
```

```
from Train t,Passenger p,T_P tp
```

```
where t.t_no=tp.t_no
```

```
and p.p_no=tp.p_no
```

```
and date1=dat
```

```
group by tp.t_no having count(tp.t_no)>=all(select count(p_no) from T_P tp
```

```
where date1=dat group by t_no);
```

```
return num;
```

```
end;
```

```
/
```

```
Function created.
```

```
-----Output For Function-----
```

```
select f1('07-JUL-16') from dual;
```

```
F1('07-JUL-16')
```

```
-----
```

```
2
```

```
-----cursor creation-----
```

```
set serveroutput on;
```

```
declare
```

```
cursor c1(dt T_P.date1%type)
```

```
is select t.t_no,t.t_name,p.p_no,p_name,addr,age from Train t,Passenger p,T_P tp
```

```
where t.t_no=tp.t_no and p.p_no=tp.p_no and date1=dt;
```

```

res c1%rowtype;

v_date T_P.date1%type;

begin

for res in c1(&v_date)

loop

dbms_output.put_line(res.t_no||' '||res.t_name||' '||res.p_name||' '||res.addr||' '||res.age);

end loop;

end;

```

-----output for cursor-----

Enter value for v_date: ' 07-JUL-16'

old 8: for res in c1(&v_date)

new 8: for res in c1(' 07-JUL-16')

2 mumbai Express yogesh pune 25

PL/SQL procedure successfully completed.

4)Consider the following entities and their relationship. Drug(d_no, d_name, company, price) Medical_store(m_no, m_name, m_city, ph_no) Relationship between Drug and Medical_Store is many-to-many with descriptive attribute quantity. Constraints: primary key, foreign key, m_name and d_name should not be null, m_city can be pune or pimpri. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1)Write a package, which consists of one procedure and one function. Pass drug number as a parameter to procedure and display details of that drug. Pass city as a parameter to a function and return total number of medical_store in given city.

2)Write a trigger that restricts insertion and updation of drug having price less than zero. (Raise user defined exception and give appropriate message)

```

create table Drug

```

```

(

```

```

d_no number(5) primary key,

```

```

d_name varchar(20) not null,

```



```
company varchar(20),

price number(5)

);

insert into Drug values(1001,'dicoliy_gel ','cipla',600);

insert into Drug values(1002,'fit_gel ','cipla',789);

insert into Drug values(1003,'crocine ','mosco',700);

create table Med_s

(

m_no number(5) primary key,

m_name varchar(20),

m_city varchar(20),

ph_no number(10)

);

insert into Med_s values(1,'raj_medical','pune',9096188754);

insert into Med_s values(2,'dipali_medical','pune',9096188755);

insert into Med_s values(3,'laxmi_medical','mumbai',9096188752);

create table D_M

(

d_no number(5) references Drug(d_no),

m_no number(5) references Med_s(m_no),

quantity number(5)

);

insert into D_M values(1001,1,5);

insert into D_M values(1001,2,4);

insert into D_M values(1002,3,8);

insert into D_M values(1003,1,8);

insert into D_M values(1003,2,9);
```

```
insert into D_M values(1003,3,10);
```

```
insert into D_M values(1001,3,5);
```

```
insert into D_M values(1002,1,6);
```

```
insert into D_M values(1002,2,7);
```

Q1-----creation of package-----

```
set serveroutput on;
```

```
create or replace package pack1
```

```
as
```

```
procedure p1(dno in number);
```

```
function f1(c Med_s.m_city%type )
```

```
return number;
```

```
end pack1;
```

```
/
```

Package created.

----- body of package-----

```
set serveroutput on;
```

```
create or replace package body pack1
```

```
as
```

```
procedure p1(dno in number)
```

```
as
```

```
a Drug%rowtype;
```

```
begin
```

```
select Drug.* into a from Drug
```

```
where d_no=dno;
```

```
dbms_output.put_line(' ');
```

```
dbms_output.put_line(' .... OUTPUT ....');
```

```
dbms_output.put_line(' ');
```



```

dbms_output.put_line('dno name company price');

dbms_output.put_line(a.d_no||' '||a.d_name||' '||a.company||' '||a.price);

end p1;

function f1(c Med_s.m_city%type)

return number

as cnt number;

begin

select count(*) into cnt from Med_s where m_city=c;

return cnt;

end f1;

end pack1;

/

```

-----call of package-----

```

SQL> set serveroutput on;

declare

n number;

name Med_s.m_city%type ;

i number;

begin

n:=&n;

pack1.p1(n);

name:= '&name';

i:=pack1.f1(name);

dbms_output.put_line(' ');

dbms_output.put_line(' ');

dbms_output.put_line(' ');

dbms_output.put_line('count of '||name||' is '||i);

end;

```

/

-----output for package& funcation-----

Enter value for n: 'raj_medical'

old 6: n:=&n;

new 6: n:='raj_medical';

Enter value for name: 'crocine '

old 8: name:= '&name';

new 8: name:= "crocine ";

name:= "crocine ";

Q2)-----creation of trigger-----

set serveroutput on;

create or replace trigger t14

before insert or update on Drug

for each row

begin

if(:new.price<0) then

raise_application_error(-20343,'drug having price less than zero.');

end if;

end;

/

-----output of trigger-----

SQL> insert into Drug values(1010,'dicoliy_gel ','cipl',-600);

insert into Drug values(1010,'dicoliy_gel ','cipl',-600)

*

ERROR at line 1:

ORA-20343: drug having price less than zero.

slip10)Consider the following entities and their relationship. University (u_no, u_name, u_city)

College (c_no, c_name, c_city, year_of_establishment) Relationship between University and College is one-to-many Constraints: primary key, foreign key, u_name and c_name should not be null. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1)Write a package, which consists of one procedure and one function. Pass university number as a parameter to procedure and display details of that university. Pass city as a parameter to a function and return total number of colleges in given city.

2)Write a cursor which will display university wise their college details. (Use parameterized cursor)

```
create table University
```

```
(
```

```
u_no number(5) primary key,
```

```
u_name varchar(20),
```

```
u_city varchar(20)
```

```
);
```

```
insert into University values(1,'solapur university','solapur');
```

```
insert into University values(2,'pune university','pune');
```

```
create table College
```

```
(
```

```
c_no number(5) primary key,
```

```
c_name varchar(20),
```

```
c_city varchar(20),
```

```
year number(5),
```

```
u_no number(5) references University(u_no)
```

```
);
```

```
insert into College values(101,'d.y.patil','pune',2000,1);
```

```
insert into College values(103,'Y.D.S','solapur',2001,2);
```

```
insert into College values(104,'D.Y.S','solapur',2002,2);
```

```
insert into College values(102,'Ramkirshna more','pune',2001,1);
```

Q1)-----creation og package-----

```
set serveroutput on;
```

```
create or replace package pack2
```

```
as
```

```
procedure p1(uno in number);
```

```
function f1(c College.c_city%type )
```

```
return number;
```

```
end pack2;
```

```
/
```

Package created.

-----body of package-----

```
set serveroutput on;
```

```
create or replace package body pack2
```

```
as
```

```
procedure p1(uno in number)
```

```
as
```

```
a University%rowtype;
```

```
begin
```

```
select University.* into a from University
```

```
where u_no=uno;
```

```
dbms_output.put_line(' ');
```

```
dbms_output.put_line(' .... OUTPUT ....');
```

```
dbms_output.put_line(' ');
```

```
dbms_output.put_line('uno university_name city');
```

```
dbms_output.put_line(a.u_no||' '||a.u_name||' '||a.u_city);
```

```
end p1;
```

```
function f1(c College.c_city%type)
```

```
return number

as cnt number;

begin

select count(*) into cnt from College where c_city=c;

return cnt;

end f1;

end pack2;

/
```

Package body created.

-----call of package-----

```
SQL> set serveroutput on;
```

```
SQL> declare
```

```
n number;
```

```
name College.c_city%type ;
```

```
i number;
```

```
begin
```

```
n:=&n;
```

```
pack2.p1(n);
```

```
name:= '&name';
```

```
i:=pack1.f1(name);
```

```
dbms_output.put_line(' ');
```

```
dbms_output.put_line(' ');
```

```
dbms_output.put_line(' ');
```

```
dbms_output.put_line('count of '||name||' is '||i);
```

```
end;
```

```
/
```

-----output of package-----

Enter value for n: 1

old 6: n:=&n;

new 6: n:=1;

Enter value for name: 'solapur university'

old 8: name:= '&name';

new 8: name:= "solapur university";

name:= "solapur university";

Q2)-----creation of cursor-----

set serveroutput on;

create or replace procedure d1(unno in number)

as

cursor c1 is select u.u_no,c.c_no,c.c_name,c.c_city,c.year from University u,College c

where u.u_no=c.u_no and c.u_no=unno;

begin

for x in c1 loop

dbms_output.put_line('university no'||x.u_no ||'college no:'||x.c_no ||'college name:'||x.c_name ||'college City:'
||x.c_city ||'year:'||x.year);

end loop;

end;

/

Procedure created.

-----output for curser-----

enter the value for unno:1

old 7: for d1('&solapur university')

new 7: for d1('1')

1 101 d.y.patil pune 2015

9) Consider the following entities and their relationship. Researcher (r_no, r_name, r_city)

Research_Paper (rp_no, rp_title, rp_subject, rp_level) Relationship between Researcher and Research_Paper is many-to-many with descriptive attribute year. Constraints:primary key, foreign key, r_name and rp_title should not be null, rp_subject can be computer, electronics or finance. rp_level can be state, national or international. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

- 1)Write a procedure which will display details of research paper of a given subject for a specified year.
- 2)Write a trigger before insert or update of each row of research_paper published after 2010 be entered into table. (Raise user defined exception and give appropriate message)

```
create table Researcher
```

```
(
```

```
  r_no number(5) primary key,
```

```
  r_name varchar(20)not null,
```

```
  r_city varchar(20)
```

```
);
```

```
insert into Researcher values(1,'yogesh','pune');
```

```
insert into Researcher values(2,'dipali','pune');
```

```
insert into Researcher values(3,'ram','mumbai');
```

```
create table Research_Paper
```

```
(
```

```
  rp_no number(5) primary key,
```

```
  rp_title varchar(20) not null,
```

```
  rp_subject varchar(20),
```

```
  rp_level varchar(20)
```

```
);
```

```
insert into Research_Paper values(101,'computer study','computer','state');
```

```
insert into Research_Paper values(102,'computer study','computer','national');
```

```
insert into Research_Paper values(103,'electronic study','electroinc','international');
```

```
insert into Research_Paper values(104,'finance study','finance','state');
```

```
insert into Research_Paper values(105,'computer study','computer','national');
insert into Research_Paper values(106,'computer study','computer','national');
insert into Research_Paper values(107,'computer study','computer','national');
insert into Research_Paper values(108,'computer study','computer','national');
```

```
create table R_RP
```

```
(
r_no number(5) references Researcher(r_no),
rp_no number(5) references Research_Paper(rp_no),
year number
);
```

```
insert into R_RP values(1,101,2015);
insert into R_RP values(1,102,2015);
insert into R_RP values(1,103,2015);
insert into R_RP values(2,101,2016);
insert into R_RP values(3,103,2015);
insert into R_RP values(2,104,2016);
insert into R_RP values(1,102,2013);
insert into R_RP values(2,102,2013);
insert into R_RP values(3,102,2013);
insert into R_RP values(1,105,2013);
insert into R_RP values(1,106,2013);
insert into R_RP values(1,107,2013);
insert into R_RP values(1,108,2013);
```

Q1)-----create procedure-----

```
Set serveroutput on;
```



```
create or replace procedure info_disp(sub varchar2,yr R_RP.year%type)
```

```
is
```

```
cursor c1 is select r.rp_no,r.rp_title,r.rp_level from Researcher r1,Research_Paper r,R_RP rrp
```

```
where r1.r_no=rrp.r_no and r.rp_no=rrp.rp_no
```

```
and r.rp_subject=sub and rrp.year=yr;
```

```
rec c1%rowtype;
```

```
begin
```

```
dbms_output.put_line(sub||' '||yr);
```

```
open c1;
```

```
loop
```

```
fetch c1 into rec;
```

```
exit when c1%notfound;
```

```
dbms_output.put_line(rec.rp_no||' '||rec.rp_title||' '||rec.rp_level);
```

```
end loop;
```

```
close c1;
```

```
end;
```

```
/
```

```
Procedure created.
```

```
-----call of procedure-----
```

```
Set serveroutput on;
```

```
declare
```

```
name_of_subject Research_Paper.rp_subject%type:='&name_of_subject';
```

```
year1 R_RP.year%type:='&year1';
```

```
begin
```

```
info_disp(name_of_subject,year1);
```

```
end;
```

```
/
```

```
-----output of procedure-----
```

Enter value for name_of_subject: 'computer'

```
old 2: name_of_subject Research_Paper.rp_subject%type:='&name_of_subject';
```

```
new 2: name_of_subject Research_Paper.rp_subject%type:="computer";
```

Enter value for year1: 2015

```
old 3: year1 R_RP.year%type:=&year1;
```

```
new 3: year1 R_RP.year%type:=2015;
```

```
name_of_subject Research_Paper.rp_subject%type:="computer";
```

Q2)-----trigger-----

Set serveroutput on;

create or replace trigger t10

before insert or update on R_RP

for each row

begin

if (:new.year<2010) then

raise_application_error(-20343,'year should be greater than current 2010');

end if;

end;

/

Trigger created.

-----output of trigger-----

```
SQL> insert into R_RP values(1,101,2009);
```

```
insert into R_RP values(1,101,2009)
```

*

ERROR at line 1:

ORA-20343: year should be greater than current 2010

Slip 12)Consider the following entities and their relationship.

Client (c_no, c_name, c_addr, birth_date)

Policy_info (p_no, p_name, maturity_amt, prem_amt, policy_term) Relationship between Client and Policy_info is many-to-many with descriptive attribute date_of_purchase. Constraints: primary key, foreign key, c_name and p_name should not be null, policy_term should be greater than zero. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1) Write a procedure which will display all policy details of given client for a given year.

2) Write a trigger which restricts insertion of policy_info having maturity amount less than premium amount. (Raise user defined exception and give appropriate message)

create table client

(

c_no number(5) primary key,

c_name varchar(20) not null,

c_addr varchar(20),

birth_date date

);

insert into client values(101,'ram','pune','13-jan-1999');

insert into client values(102,'sham','mumbai','03-may-1997');

insert into client values(103,'sia','pune','16-jan-1998');

create table policy_info

(

p_no number(5) primary key,

p_name varchar(20),

maturity_amt number(10),

prem_amt number(10),

policy_term number(10) constraint chkpolicy_info check(policy_term>0)

);

```
-----  
  
insert into policy_info values(201,'insurance',1000,2000,5);  
  
insert into policy_info values(202,'insurance' ,2000,2000,4);  
  
insert into policy_info values(203,'insurance',1000,3000,3);  
  
-----
```

```
create table cp_info  
  
(  
  
c_no number(5) references client(c_no),  
  
p_no number(5) references policy_info(p_no),  
  
date_of_purchase date  
  
);  
  
-----
```

```
insert into cp_info values(101,201,'3-may-2020');  
  
insert into cp_info values(103,203,'23-jan-2019');  
  
insert into cp_info values(102,202,'24-may-2020');
```

Q1)-----create procedure-----

```
set serveroutput on;  
  
create or replace procedure p2(n in varchar)as cursor c1 is selet c_name,  
  
p_name,maturity_amt,prem_amt,policy_term,date_of_purchase  
  
from client,policy_info,cp_info  
  
where client.c_no=cp_info.c_no and  
  
policy_info.p_no=cp_info.p_no and  
  
c_name='ram';  
  
c c1%rowtype;  
  
begin  
  
open c1;  
  
dbms_output.put_line('c_name' || "|| 'p_name' || "|| 'maturity_amt' || "|| 'prem_amt' || "|| 'policy_term');
```

```
loop

fetch c1 into c;

exit when c1% notfound;

if(c.year=n)then

dbms_output.put_line('c_name'||"|| 'p_name'||"|| 'maturity_amt'||"|| 'prem_amt' ||"|| 'policy_term');

end if;

end loop;

close c1;

end;
```

/

-----call for procedure-----

begin

end

/

-----output of procedure-----

c_name p_name maturity_amt prem_amt policy_term

ram insurance 1000 2000 5

PL/SQL procedure successfully completed.

Q2)-----create trigger-----

set serveroutput on;

create or replace trigger t6

before insert or update on policy_info

for each row

Begin

if(:new.maturity_amt <:new.prem_amt) then

raise_application_error(-20343,'maturity amount less than premium amount');

End if;

End;

/

trigger created.

-----output of trigger-----

SQL>insert into policy_info values(204,'insurance',1000,3000,5);

insert into policy_info values(204,'insurance',1000,3000,5);

*

ERROR at line 1:

ORA-20343: maturity amount less than premium amount.

Slip13)Consider the following entities and their relationship. Company (c_no, c_name, c_city, c_share_value) Person (p_no, p_name, p_city, p_ph_no) Relationship between Company and Person is many-to-many with descriptive attribute no_of_shares. Constraints:primary key, foreign key, c_name and p_name should not be null, no_of_shares should be greater than zero. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1)Write a procedure which will display names of person who are shareholder of the given company.

2)Write a trigger which get activated when share value of company become less than Rs. 10.

(Raise user defined exception and give appropriate message)

create table Company

(

c_no number(5) primary key,

c_name varchar(20) not null,

c_city varchar(20),

c_share_value number(5)

);

insert into Company values(101,'infosys','pune',19);


```
insert into Company values(102,'Wipro','pune',20);
```

```
insert into Company values(103,'vinsys','mumbai',11);
```

```
create table Person
```

```
(
```

```
p_no number(5) primary key,
```

```
p_name varchar(20) not null,
```

```
p_city varchar(20),
```

```
p_ph_no number(10)
```

```
);
```

```
insert into Person values(1,'yogesh','pune',9096188754);
```

```
insert into Person values(2,'dipali','pune',9096188755);
```

```
insert into Person values(3,'amol','mumbai',9096188764);
```

```
create table CO_P
```

```
(
```

```
c_no number(5) references Company(c_no),
```

```
p_no number(5) references Person(p_no),
```

```
no_of_shares number(5)
```

```
);
```

```
insert into CO_P values(101,1,5000);
```

```
insert into CO_P values(101,2,4000);
```

```
insert into CO_P values(102,1,6000);
```

```
insert into CO_P values(103,3,5000);
```

Q1)-----creation of procedure-----

```
set serveroutput on;
```

```
create or replace procedure p1
```

```
as
```

```
cursor c1 is select c_name from Company;
```

```
cursor c2(cname in Company.c_name%type) is select distinct p_name from Company c,Person p,CO_P cp
```

```
where c.c_no=cp.c_no and p.p_no=cp.p_no and c_name=cname;
```

```
r1 c1%rowtype;
```

```
r2 c2%rowtype;
```

```
begin
```

```
open c1;
```

```
loop
```

```
fetch c1 into r1;
```

```
exit when c1%notfound;
```

```
dbms_output.put_line(r1.c_name);
```

```
open c2(r1.c_name);
```

```
loop
```

```
fetch c2 into r2;
```

```
exit when c2%notfound;
```

```
dbms_output.put_line(r2.p_name);
```

```
end loop;
```

```
close c2;
```

```
end loop;
```

```
close c1;
```

```
end;
```

```
/
```

```
procedure created.
```

```
-----call for procedure-----
```

```
declare
```


begin

p1;

end;

/-----output of procedure-----

infosys

yogesh

dipali

wipro

yogesh

vinsys

amol

PL/SQL procedure successfully completed.

Q2)-----create a trigger-----

set serveroutput on;

create or replace trigger t10

before insert or update on Company

for each row

begin

if (:new.c_share_value<10) then

raise_application_error(-20343,'share of value should be greater than 10');

end if;

end;

/

Trigger created.

-----output of trigger-----

SQL>insert into Company values(104,'tcs','pune',8);

insert into Company values(104,'tcs','pune',8);

*

Error at line 1:

ORA-20343:share of value should be greater than 10.

Slip-14)Consider the following entities and their relationship. Donor (donor_no, donor_name, city)

Blood_donation_detail (bd_no, blood_group, qty, date_of_collection) Relationship between Donor and Blood_donation_detail is one-to-many. Constraints:primary key, foreign key, donor_name should not be null, blood_group can be A+, A-, B+, B-, AB+, AB-, O+, O- Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

1)Write a procedure which will display blood group wise total amount of quantity of blood available.

2)Write a trigger that restricts insertion of blood_donation_details having quantity greater than 300ml.

(Raise user defined exception and give appropriate message)

create table Donor

(

donor_no number(5) primary key,

donor_name varchar(20) not null,

city varchar(20)

);

insert into Donor values(1,'yogesh','pune');

insert into Donor values(2,'dipali','pune');

insert into Donor values(3,'amol','mumbai');

create table Blood_donation_detail

(

bd_no number(5) primary key,

blood_group varchar(20),

qty number(5),

date_of_collection date,

donor_no number references Donor(donor_no)

);

insert into Blood_donation_detail values(1001,'A+',250,'02-aug-2016',1);

insert into Blood_donation_detail values(1002,'B+',250,'02-aug-2016',2);

insert into Blood_donation_detail values(1003,'AB+',250,'02-aug-2016',3);

insert into Blood_donation_detail values(1004,'A+',300,'04-jan-2016',1);

insert into Blood_donation_detail values(1005,'B+',250,'04-jan-2016',2);

Q1)-----create procedure-----

set serveroutput on;

create or replace procedure p3

is

cursor c1 is select blood_group,qty from Blood_donation_detail ;

begin

for a in c1 loop

dbms_output.put_line(a.blood_group||' '||a.qty);

end loop;

end;

/

procedure created.

-----call of procedure-----

declare

begin

p3;

end;

/

-----output of procedure-----

A+ 250

B+ 250

AB+ 250

A+ 300

B+ 250

PL/SQL procedure successfully completed.

Q2)-----create trigger-----

set serveroutput on;

create or replace trigger t22

before insert on blood_donation_detail

for each row

begin

if(:new.qty<=300) then

raise_application_error(-20343,'qty should be enter greater than 300 ml');

end if;

end;

/

trigger created.

-----output of trigger-----

SQL>insert into Blood_donation_detail values(1006,'A+',250,'02-aug-2016',1);

insert into Blood_donation_detail values(1006,'A+',250,'02-aug-2016',1);

*

ERROR at line 1:

ORA-20343: qty should be enter greater than 300 ml.

slip 15) Consider the following entities and their relationship. Person (p_no, p_name, p_addr) Investment (inv_no, inv_name, inv_date, inv_amt) Relationship between Person and Investment is one-to-many. Constraints: primary key, foreign key, p_name and inv_name should not be null, inv_amt should be greater than 10000. Create a RDB in 3NF and write PL/SQL blocks in Oracle for the following:

- 1) Write a procedure which will display details of person, made investment on given date.
- 2) Write a trigger that restricts insertion or updation of investment having inv_date greater than current date.
(Raise user defined exception and give appropriate message)

create table person1

(

p_no number(10) primary key,

p_name varchar(15) not null,

p_addr varchar(15)

);

insert into person1 values(1,'Abhi','pimpri');

insert into person1 values(2,'Ajay','pune');

create table investment

(

inv_no number(10) primary key,

inv_name varchar(15) not null,

inv_date date,

inv_amt number(10),

p_no number(10) references person1(p_no),

check(inv_amt>10000)

);

insert into investment values(101,'Ajay','21-dec-2021',20000,1);

```
insert into investment values(102,'Vijay','21-dec-2021',20000,2);
```

Q1)-----create procedure-----

```
set serveroutput on;
```

```
create or replace procedure p2(dt in date)
```

```
as cursor c1 is select p.p_no,p.p_name,p.p_addr from person1 p,investment
```

```
where p.p_no=i.p_no and inv_date=dt;
```

```
begin
```

```
for x in c1 loop
```

```
dbms_output.put_line(x.p_no||' '||x.p_name||' '||x.p_addr);
```

```
end loop;
```

```
end;
```

```
/
```

```
procedure created.
```

-----call for procedure-----

```
declare
```

```
begin
```

```
p1('21-dec-2021');
```

```
end;
```

```
/
```

-----output of procedure-----

```
1 Abhi pimpri
```

```
2 Ajay pune
```

```
PL/SQL procedure successfully completed.
```

Q2)-----create trigger-----

```
set serveroutput on;
```

```
create or replace trigger t8
```

```
before insert or update on investment
```


for each row

Begin

if(:new.inv_date < sysdate) then

raise_application_error(-20343,'inv_date should be greater than sysdate');

End if;

End;

/

trigger created.

-----Output of trigger-----

SQL>insert into investment values(103,'sanjay','2-feb-2022',20000,4);

insert into investment values(103,'sanjay','2-feb-2022',20000,4)

*

ERROR at line 1:

ORA-20343: inv_date should be greater than sysdate.

